Continued Fractions

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1 History

Continued fractions were often times used off-handedly in the last 2000 years, but the foundations of continued fractions were not established until late 1600's, early 1700's by John Wallis, see a fuller account in the Mathematics Archives [1]. Below is an except from the Mathematics Archives [1].

> In his book Opera Mathematica (1695) Wallis laid some basic groundwork for continued fractions. He explained how to compute the nth convergent and discovered some now familiar properties of convergents. It was also in this work that the term "continued fraction" was first used.
>
> The Dutch mathematician and astronomer Christiaan Huygens (1629-1695) was the first to demonstrate a practical application of continued fractions. He wrote a paper explaining how to use the convergents of a continued fraction to find the best rational approximations for gear ratios. These approximations enabled him to pick the gears with the correct number of teeth. His work was motivated impart by his desire to build a mechanical planetarium.
>
> While the work of Wallis and Huygens began the work on continued fractions, the field of continued fractions began to flourish when Leonard Euler (1707-1783), Johan Heinrich Lambert (1728-1777), and Joseph Louis Lagrange (1736-1813) embraced the topic. Euler laid down much of the modern theory in his work De Fractionlous Continious (1737). He showed that every rational can be expressed as a terminating simple continued fraction. He also provided an expression for e in continued fraction form. He used this expression to show that e and e^2 are irrational. He also demonstrated how to go from a series to a continued fraction representation of the series, and conversely.
>
> Lambert generalized Euler's work on e to show that both e^x and $\tan(x)$ are irrational if x is rational. Lagrange used continued fractions to find the value of irrational roots. He also proved that a real root of a quadratic irrational is a periodic continued fraction.
>
> The nineteenth century can probably be described as the golden age of continued fractions. As Claude Brezinski writes in History of Continued Fractions and Padre Approximations, "the nineteenth

century can be said to be popular period for continued fractions." It was a time in which "the subject was known to every mathematician." As a result, there was an explosion of growth within this field. The theory concerning continued fractions was significantly developed, especially that concerning the convergents. Also studied were continued fractions with complex variables as terms. Some of the more prominent mathematicians to make contributions to this field include Jacobi, Perron, Hermite, Gauss, Cauchy, and Stieljes. By the beginning of the 20th century, the discipline had greatly advanced from the initial work of Wallis.

2 Definition

Definition 2.1 (Finite General Continued Fraction)**.** *A **finite general** continued fraction is defined as:*

$$a_0 + \cfrac{b_1}{a_1 + \cfrac{b_2}{a_2 + \cfrac{b_3}{\ddots + \cfrac{b_n}{a_n}}}}$$

If we allow the number of levels to go to infinity we get a general continued fraction.

Definition 2.2 (General Continued Fraction)**.** *A **general** continued fraction is defined as:*

$$a_0 + \cfrac{b_1}{a_1 + \cfrac{b_2}{a_2 + \cfrac{b_3}{a_3 + \ddots}}}$$

Definition 2.3 (Finite Standard Continued Fraction)**.** *A **finite standard** continued fraction is defined as:*

$$[a_0; a_1, a_2, \cdots, a_n] = a_0 + \cfrac{1}{a_1 + \cfrac{1}{a_2 + \cfrac{1}{\ddots + \cfrac{1}{a_n}}}}$$

If we allow the number of levels to go to infinity we get a standard continued fraction.

Definition 2.4 (Standard Continued Fraction)**.** *A **standard** continued fraction is defined as:*

$$a_0 + \cfrac{1}{a_1 + \cfrac{1}{a_2 + \cfrac{1}{a_3 + \ddots}}}$$

This form is also referred to as the simple continued fraction or regular continued fraction, or the canonical form.

2.1 Rational Examples

Every rational number has a unique representation in the standard continued fraction form. Moreover, every rational number can be represented using only the finite continued fraction form. Below are a few examples.

$$\frac{1}{8} = [0;8] = \mathbf{0} + \frac{1}{\mathbf{8}}$$

$$\frac{2}{5} = [0;2,2] = \mathbf{0} + \cfrac{1}{\mathbf{2} + \cfrac{1}{\mathbf{2}}}$$

$$\frac{163}{7} = [23;3,2] = \mathbf{23} + \cfrac{1}{\mathbf{3} + \cfrac{1}{\mathbf{2}}}$$

$$\frac{-13}{5} = [-3;2,2] = \mathbf{-3} + \cfrac{1}{\mathbf{2} + \cfrac{1}{\mathbf{2}}}$$

2.2 Generation of continued fractions

The simple continued fraction representation of a number is both easy to verify and easy to generate. Here we go through two common ways of generating the simple continued fraction representation of a number.

2.2.1 Rational numbers

Given a rational number p/q we can use Euclid's algorithm for determining the greatest common divisor (GCD) between the numbers p and q.
Using the example of $\frac{163}{7}$ as listed above we get the following steps:

See how many times 7 goes into 163 evenly

$$163 = \mathbf{23} \cdot 7 + \mathbf{2}$$

See how many times 2 goes into 7 evenly

$$7 = \mathbf{3} \cdot 2 + \mathbf{1}$$

See how many times 1 goes into 2 evenly

$$2 = \mathbf{2} \cdot 1 + \mathbf{0}$$

Notice that the resulting divisor values exactly match the corresponding simple continued fraction representation of $\frac{163}{7}$ as shown above. In particular

$$\frac{163}{7} = [23;3,2] = \mathbf{23} + \cfrac{1}{\mathbf{3} + \cfrac{1}{\mathbf{2}}}$$

2.2.2 General real numbers

We can apply the same reasoning as for the rational numbers to general real numbers.
Using $\sqrt{2} = 1.4142135623730951\ldots$ as an example.

Remove the largest integer from value

$$\sqrt{2} = \mathbf{1} + 0.4142135623730951\ldots$$

Take the reciprocal of the result and remove the largest integer from that

$$1/0.4142135623730951\cdots = 2.414213562373095 = \mathbf{2} + .414213562373095$$

Repeat the process

$$1/0.414213562373095\cdots = 2.41421356237309 = \mathbf{2} + .414213562373095$$

These calculations seem to imply that

$$\sqrt{2} = [1; 2, 2, 2, \ldots] = \mathbf{1} + \cfrac{1}{\mathbf{2} + \cfrac{1}{\mathbf{2} + \cfrac{1}{\mathbf{2} + \ddots}}}$$

We will see in a later section that this is the exact continued fraction representation of $\sqrt{2}$.

Moreover, we will show that in general if $\sqrt{n}$ is not an integer then it will always have a periodic repeating infinite continued fraction representation.

The algorithm presented here will work for every number as long as a decimal representation is available for said number.

3 Useful lemmas

Definition 3.1 (Convergents)**.** *A convergent $\frac{p_n}{q_n}$ of a simple continued fraction $[a_0; a_1, a_2, a_3, \ldots]$ is defined as:*

$$\frac{p_n}{q_n} = [a_0; a_1, a_2, a_3, \ldots, a_n]$$

For reasons that will become apparent later we define $p_{-1} = 1, p_{-2} = 0$ and $q_{-1} = 0, q_{-2} = 1$.

Lemma 3.2 (Recurrence Relation)**.** *For $\frac{p_n}{q_n}$ the nth convergent of $[a_0; a_1, a_2, a_3, \ldots]$.*

We can find p_n, q_n using the previous convergents through the following recurrence relation:

$$\frac{p_n}{q_n} = \frac{a_n \cdot p_{n-1} + p_{n-2}}{a_n \cdot q_{n-1} + q_{n-2}}$$

Proof.

$$\frac{p_0}{q_0} = a_0 = \frac{a_0 \cdot 1 + 0}{a_0 \cdot 0 + 1} = \frac{a_0 \cdot p_{-1} + p_{-2}}{a_0 \cdot q_{-1} + q_{-2}}$$
$$\frac{p_1}{q_1} = a_0 + \frac{1}{a_1} = \frac{a_0 a_1 + 1}{a_1} = \frac{a_1 \cdot a_0 + 1}{a_1 \cdot 1 + 0}$$
$$\frac{p_2}{q_2} = a_0 + \frac{1}{a_1 + \frac{1}{a_2}} = a_0 + \frac{a_2}{a_1 a_2 + 1} = \frac{a_0 a_1 a_2 + a_0 + a_2}{a_1 a_2 + 1} = \frac{a_2 \cdot (a_0 a_1 + 1) + a_0}{a_2 \cdot a_1 + 1}$$

Assume true for k and prove for $k+1$:

$$\frac{p_{k+1}}{q_{k+1}} = a_0 + \cfrac{1}{a_1 + \cfrac{1}{\ddots + \cfrac{1}{a_k + \cfrac{1}{a_{k+1}}}}}$$
$$\frac{p_{k+1}}{q_{k+1}} = a_0 + \cfrac{1}{a_1 + \cfrac{1}{\ddots + \cfrac{1}{\cfrac{a_k a_{k+1} + 1}{a_{k+1}}}}}$$

Let $a_k^{'} = \frac{a_k a_{k+1} + 1}{a_{k+1}}$ and use the assumption for $\frac{p_k}{q_k}$:

$$\frac{p_{k+1}}{q_{k+1}} = \frac{\frac{a_k a_{k+1} + 1}{a_{k+1}} \cdot p_{k-1} + p_{k-2}}{\frac{a_k a_{k+1} + 1}{a_{k+1}} \cdot q_{k-1} + q_{k-2}}$$
$$\frac{p_{k+1}}{q_{k+1}} = \frac{(a_k a_{k+1} + 1) \cdot p_{k-1} + a_{k+1} p_{k-2}}{(a_k a_{k+1} + 1) \cdot q_{k-1} + a_{k+1} q_{k-2}}$$
$$\frac{p_{k+1}}{q_{k+1}} = \frac{a_{k+1}(a_k p_{k-1} + p_{k-2}) + p_{k-1}}{a_{k+1}(a_k q_{k-1} + q_{k-2}) + q_{k-1}}$$

Again using the recursive assumption for p_k and q_k we finish the inductive proof:

$$\frac{p_{k+1}}{q_{k+1}} = \frac{a_{k+1} p_k + p_{k-1}}{a_{k+1} q_k + q_{k-1}}$$

□

Lemma 3.3 (Determinant Identity)**.** *For $\frac{p_n}{q_n}$, the nth convergent of $[a_0; a_1, a_2, a_3, \ldots]$:*

$$p_{n-1} \cdot q_n - p_n \cdot q_{n-1} = (-1)^n$$

Proof. We proceed by verifying the base cases and then using mathematical induction on n.

Base Cases:

$$n = 0: \quad p_{-1} \cdot q_0 - p_0 \cdot q_{-1} = 1 \cdot 1 - a_0 \cdot 0 = 1 = (-1)^0,$$
$$n = 1: \quad p_0 \cdot q_1 - p_1 \cdot q_0 = a_0 \cdot a_1 - (a_0 a_1 + 1) \cdot 1 = -1 = (-1)^1,$$

Induction Hypothesis: Assume that for k, the statement holds:

$$p_{k-1} \cdot q_k - p_k \cdot q_{k-1} = (-1)^k.$$

Inductive Step: We need to prove the statement for $k+1$:

$$\begin{aligned} p_k \cdot q_{k+1} - p_{k+1} \cdot q_k &= p_k \cdot (a_{k+1}q_k + q_{k-1}) - (a_{k+1}p_k + p_{k-1}) \cdot q_k \\ &= (p_k \cdot q_{k-1} - p_{k-1} \cdot q_k) \\ &= -(p_{k-1} \cdot q_k - p_k \cdot q_{k-1}) = -(-1)^k = (-1)^{k+1}. \end{aligned}$$

Therefore, $p_k \cdot q_{k+1} - p_{k+1} \cdot q_k = (-1)^{k+1}$, completing the induction step. □

Lemma 3.4 (Difference Identity)**.** *For $\frac{p_n}{q_n}$ the nth convergent of $[a_0; a_1, a_2, a_3, \dots]$.*
We can define a closed form for the difference of two sequential convergents:

$$\frac{p_{n-1}}{q_{n-1}} - \frac{p_n}{q_n} = \frac{(-1)^n}{q_n \cdot q_{n-1}}$$

Proof.

$$\frac{p_{n-1}}{q_{n-1}} - \frac{p_n}{q_n} = \frac{p_{n-1} \cdot q_n - p_n \cdot q_{n-1}}{q_n q_{n-1}}$$

Using Lemma 3.3 we have

$$\frac{p_{n-1}}{q_{n-1}} - \frac{p_n}{q_n} = \frac{(-1)^n}{q_n q_{n-1}}$$

□

4 Relation to alternating Series

Theorem 4.1 (Series to Continued Fraction)**.**

$$\sum_{k=1}^{n} \frac{(-1)^{k-1}}{a_k} = \cfrac{1}{a_1 + \cfrac{a_1^2}{a_2 - a_1 + \cfrac{a_2^2}{a_3 - a_2 \cfrac{a_3^2}{\cdots + \cfrac{a_{n-1}^2}{a_n - a_{n-1}}}}}}$$

Proof. **Proof by induction**
Base Case n=1:

$$\sum_{k=1}^{1} \frac{(-1)^{k-1}}{a_k} = \frac{1}{a_1} = \frac{1}{a_1}$$

Base Case n=2:

$$\begin{aligned}\sum_{k=1}^{2}\frac{(-1)^{k-1}}{a_k} &= \frac{1}{a_1}-\frac{1}{a_2}\\ &= \frac{a_2-a_1}{a_1\cdot a_2}\\ &= \frac{1}{\frac{a_1\cdot a_2}{a_2-a_1}}\\ &= \frac{1}{a_1+\frac{a_1^2}{a_2-a_1}}\end{aligned}$$

Inductive Hypothesis: Assume the statement is true for n:

$$\sum_{k=1}^{n}\frac{(-1)^{k-1}}{a_k} = \cfrac{1}{a_1+\cfrac{a_1^2}{a_2-a_1+\cfrac{a_2^2}{a_3-a_2\cfrac{a_3^2}{\ddots+\cfrac{a_{n-1}^2}{a_n-a_{n-1}}}}}}$$

Inductive Step: Prove for $n+1$.

The idea of the proof is to re-write the summation for $n+1$ as a summation over n terms by obtaining a new $\widehat{a_n}$ term.

$$\begin{aligned}\sum_{k=1}^{n+1}\frac{(-1)^{k-1}}{a_k} =&\frac{1}{a_1}-\frac{1}{a_2}+\frac{1}{a_3}-\cdots+\frac{(-1)^{n-1}}{a_n}+\frac{(-1)^n}{a_{n+1}}\\ =&\frac{1}{a_1}-\frac{1}{a_2}+\frac{1}{a_3}-\cdots+(-1)^{n-1}\cdot\left(\frac{1}{a_n}-\frac{1}{a_{n+1}}\right)\\ =&\frac{1}{a_1}-\frac{1}{a_2}+\frac{1}{a_3}-\cdots+(-1)^{n-1}\cdot\left(\frac{a_{n+1}-a_n}{a_n\cdot a_{n+1}}\right)\\ =&\frac{1}{a_1}-\frac{1}{a_2}+\frac{1}{a_3}-\cdots+(-1)^{n-1}\cdot\left(\frac{1}{\frac{a_n\cdot a_{n+1}}{a_{n+1}-a_n}}\right)\\ =&\frac{1}{a_1}-\frac{1}{a_2}+\frac{1}{a_3}-\cdots+(-1)^{n-1}\frac{1}{\widehat{a_n}}\end{aligned}$$

Note that

$$\widehat{a_n}=\frac{a_n\cdot a_{n+1}}{a_{n+1}-a_n}=\frac{(a_{n+1}-a_n)\cdot a_n+a_n^2}{a_{n+1}-a_n}=a_n+\frac{a_n^2}{a_{n+1}-a_n}$$

Hence, plugging in this new $\widehat{a_n}$ into our inductive hypothesis we get:

$$\sum_{k=1}^{n+1} \frac{(-1)^{k-1}}{a_k} = \cfrac{1}{a_1 + \cfrac{a_1^2}{a_2 - a_1 + \cfrac{a_2^2}{a_3 - a_2 + \cfrac{a_3^2}{\ddots + \cfrac{a_{n-1}^2}{\widehat{a_n} - a_{n-1}}}}}}$$

$$= \cfrac{1}{a_1 + \cfrac{a_1^2}{a_2 - a_1 + \cfrac{a_2^2}{a_3 - a_2 + \cfrac{a_3^2}{\ddots + \cfrac{a_{n-1}^2}{\left(a_n + \frac{a_n^2}{a_{n+1} - a_n}\right) - a_{n-1}}}}}}$$

$$= \cfrac{1}{a_1 + \cfrac{a_1^2}{a_2 - a_1 + \cfrac{a_2^2}{a_3 - a_2 + \cfrac{a_3^2}{\ddots + \cfrac{a_{n-1}^2}{a_n - a_{n-1} + \frac{a_n^2}{a_{n+1} - a_n}}}}}}$$

Which proves the theorem for $n+1$ since this matches the form we were expecting for $n+1$. □

Theorem 4.2 (Euler's Continued Fraction Formula)**.** *https://en.wikipedia.org/wiki/Euler%27s_continued_fraction_formula*

$$\sum_{k=0}^{n} a_0 a_1 \cdots a_k = \cfrac{a_0}{1 - \cfrac{a_1}{1 + a_1 - \cfrac{a_2}{1 + a_2 - \cfrac{a_3}{\ddots + 1 + a_{n-1} - \cfrac{a_n}{1 + a_n}}}}}$$

Proof. We proceed by induction on n.

Base case $(n = 0)$**.**

$$\sum_{k=0}^{0} a_0 a_1 \cdots a_k = a_0, \qquad \frac{a_0}{1} = a_0.$$

Base case $(n = 1)$**.**

$$\text{LHS} = a_0 + a_0 a_1 = a_0(1 + a_1),$$

$$\text{RHS} = \frac{a_0}{1 - \dfrac{a_1}{1 + a_1}} = \frac{a_0(1 + a_1)}{1 + a_1 - a_1} = a_0(1 + a_1).$$

Inductive hypothesis. Assume for some $k \geq 1$ that

$$\sum_{j=0}^{k} a_0 a_1 \cdots a_j = \cfrac{a_0}{1 - \cfrac{a_1}{1 + a_1 - \cfrac{a_2}{\ddots - \cfrac{a_k}{1 + a_k}}}}.$$

Inductive step. We show the formula holds for $k+1$.

Let

$$S_k := \sum_{j=0}^{k} a_0 a_1 \cdots a_j.$$

Then

$$S_{k+1} = S_k + a_0 a_1 \cdots a_{k+1} = S_k \left(1 + a_{k+1} \frac{a_0 a_1 \cdots a_k}{S_k}\right).$$

From the inductive hypothesis, the ratio

$$\frac{a_0 a_1 \cdots a_k}{S_k}$$

is exactly the value of the *last convergent* of the continued fraction, namely

$$\frac{a_k}{1 + a_k}.$$

Hence

$$S_{k+1} = S_k \left(1 + \frac{a_k a_{k+1}}{1 + a_k}\right).$$

Now observe that appending one more level to the continued fraction replaces the final denominator

$$1 + a_k \quad \text{by} \quad 1 + a_k - \frac{a_{k+1}}{1 + a_{k+1}}.$$

Indeed,

$$1 + a_k - \frac{a_{k+1}}{1 + a_{k+1}} = \frac{(1 + a_k)(1 + a_{k+1}) - a_{k+1}}{1 + a_{k+1}} = \frac{1 + a_k + a_{k+1} + a_k a_{k+1}}{1 + a_{k+1}}.$$

This produces exactly the multiplicative factor

$$1 + \frac{a_k a_{k+1}}{1 + a_k}$$

in the numerator, while preserving the continued fraction structure above it.

Therefore,

$$S_{k+1} = \cfrac{a_0}{1 - \cfrac{a_1}{1 + a_1 - \cfrac{a_2}{\ddots - \cfrac{a_{k+1}}{1 + a_{k+1}}}}}.$$

This completes the induction. □

Theorem 4.3 (Infinite Product Alternatives)**.**

$$\sum_{k=1}^{n} \frac{(-1)^{k-1}}{a_1 a_2 \cdots a_k} = \cfrac{1}{a_1 + \cfrac{a_1}{a_2 - 1 + \cfrac{a_2}{a_3 - 1 + \cfrac{a_3}{\ddots + a_{n-1} - 1 + \cfrac{a_{n-1}}{a_n - 1}}}}}$$

Proof. **Proof by induction**

We proceed by induction on n.

Base Case: For $n = 1$, the left-hand side of the equation is:

$$\frac{(-1)^{1-1}}{a_1} = \frac{1}{a_1}$$

And the right-hand side for $n = 1$ is:

$$\frac{1}{a_1}$$

Thus, the equation holds for $n = 1$.

Base Case: For $n = 2$, the left-hand side of the equation is:

$$\begin{aligned}\frac{1}{a_1} - \frac{1}{a_1 a_2} &= \frac{a_2 - 1}{a_1 a_2} = \frac{1}{\frac{a_1 a_2}{a_2 - 1}} \\ &= \frac{1}{\frac{a_1(a_2-1)+a_1}{a_2-1}} \\ &= \frac{1}{a_1 + \frac{a_1}{a_2 - 1}}\end{aligned}$$

And the right-hand side for $n = 2$ is:

$$\frac{1}{a_1 + \frac{a_1}{a_2 - 1}}$$

Thus, the equation holds for $n = 2$.

Inductive Step: Assume that the equation holds for some $n = K$, that is:

$$\sum_{k=1}^{K} \frac{(-1)^{k-1}}{a_1 a_2 \cdots a_k} = \cfrac{1}{a_1 + \cfrac{a_1}{a_2 - 1 + \cfrac{a_2}{a_3 - 1 + \ddots + \cfrac{a_{K-2}}{a_{K-1} - 1 + \cfrac{a_{K-1}}{a_K - 1}}}}}$$

We want to prove that the equation holds for $n = K + 1$, the left-hand side becomes:

$$\begin{aligned}
\sum_{k=1}^{K+1} \frac{(-1)^{k-1}}{a_1 a_2 \cdots a_k} &= \sum_{k=1}^{K-1} \frac{(-1)^{k-1}}{a_1 a_2 \cdots a_k} + \frac{(-1)^{K-1}}{a_1 a_2 \cdots a_K} - \frac{(-1)^{K-1}}{a_1 a_2 \cdots a_{K+1}} \\
&= \sum_{k=1}^{K-1} \frac{(-1)^{k-1}}{a_1 a_2 \cdots a_k} + \frac{(-1)^{K-1}}{a_1 a_2 \cdots a_{K-1}} \cdot \left(\frac{1}{a_K} - \frac{1}{a_K a_{K+1}} \right) \\
&= \sum_{k=1}^{K-1} \frac{(-1)^{k-1}}{a_1 a_2 \cdots a_k} + \frac{(-1)^{K-1}}{a_1 a_2 \cdots a_{K-1}} \cdot \left(\frac{a_{K+1} - 1}{a_K a_{K+1}} \right)
\end{aligned}$$

Define $\widehat{a_K} = \frac{a_K a_{K+1}}{a_{K+1}-1}$

$$= \sum_{k=1}^{K} \frac{(-1)^{k-1}}{a_1 a_2 \cdots a_{K-1} \widehat{a_K}}$$

$$= \cfrac{1}{a_1 + \cfrac{a_1}{a_2 - 1 + \cfrac{a_2}{a_3 - 1 + \ddots + \cfrac{a_{K-2}}{a_{K-1} - 1 + \cfrac{a_{K-1}}{\widehat{a_K} - 1}}}}}$$

$$= \cfrac{1}{a_1 + \cfrac{a_1}{a_2 - 1 + \cfrac{a_2}{a_3 - 1 + \ddots + \cfrac{a_{K-2}}{a_{K-1} - 1 + \cfrac{a_{K-1}}{\frac{a_K a_{K+1}}{a_{K+1}-1} - 1}}}}}$$

$$= \cfrac{1}{a_1 + \cfrac{a_1}{a_2 - 1 + \cfrac{a_2}{a_3 - 1 + \ddots + \cfrac{a_{K-2}}{a_{K-1} - 1 + \cfrac{a_{K-1}}{\frac{a_K a_{K+1} - a_{K+1} + 1}{a_{K+1}-1}}}}}}$$

$$= \cfrac{1}{a_1 + \cfrac{a_1}{a_2 - 1 + \cfrac{a_2}{a_3 - 1 + \ddots + \cfrac{a_{K-2}}{a_{K-1} - 1 + \cfrac{a_{K-1}}{\frac{(a_K - 1)\cdot(a_{K+1}-1) + a_K}{a_{K+1}-1}}}}}}$$

$$= \cfrac{1}{a_1 + \cfrac{a_1}{a_2 - 1 + \cfrac{a_2}{a_3 - 1 + \ddots + \cfrac{a_{K-2}}{a_{K-1} - 1 + \cfrac{a_{K-1}}{a_K - 1 + \frac{a_K}{a_{K+1}-1}}}}}}$$

□

4.1 Example $\ln(2)$

Lemma 4.4 (Taylor Series for ln)**.**

$$\ln(1+x) = \sum_{k=1}^{\infty} \frac{(-1)^{k+1} \cdot x^k}{k}$$

Proof. This can be shown using the Taylor Series for $\ln(x+1)$ centred around $a = 0$. See Darboux document for a proof of the Taylor Series.

Note that

$$\frac{\partial^n \big(\ln(x+1)\big)}{\partial x^n}|_{x=0} = (-1)^{n+1} \cdot (n-1)!$$

Thus plugging this into Taylor's equation we get:

$$\begin{aligned}
\ln(1+x) &= \sum_{m=0}^{\infty} \frac{(x-a)^m f^{(m)}(a)}{m!} \\
&= \sum_{m=0}^{\infty} \frac{(-1)^{m+1} \cdot x^m (m-1)!}{m!} \\
&= \sum_{m=0}^{\infty} \frac{(-1)^{m+1} \cdot x^m}{m}
\end{aligned}$$

□

Lemma 4.5 (Continued Fraction for $\ln(2)$)**.**

$$\ln(2) = \sum_{k=1}^{n} \frac{(-1)^{k+1}}{k} = \cfrac{1}{1 + \cfrac{1^2}{1 + \cfrac{2^2}{1 + \cfrac{3^2}{1 + \ddots}}}}$$

Proof. Plugging in $x = 1$ to the Taylor expansion of $\ln(x+1)$ and using Theorem 4.1 we obtain the result. □

4.2 Example $\frac{\pi}{4}$

Lemma 4.6 (Taylor Series for arctan)**.**

$$\arctan(x) = \sum_{k=1}^{\infty} \frac{(-1)^{k+1} \cdot x^{2k-1}}{2k-1}$$

Proof. This can be shown using the Taylor Series for $\arctan(x)$ centred around $a = 0$. See Darboux document for a proof of the Taylor Series. □

Lemma 4.7 (Continued Fraction for $\pi/4$)**.**

$$\frac{\pi}{4} = \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{2k-1} = \cfrac{1}{1 + \cfrac{1^2}{2 + \cfrac{3^2}{2 + \cfrac{5^2}{2 + \ddots}}}}$$

Proof. Plugging in $x = 1$ to the Taylor expansion of $\arctan(x) = \arctan(1) = \frac{\pi}{4}$ and using Theorem 4.1 we obtain the result. □

5 Square root numbers

Theorem 5.1 (Square Root CF)**.** *For any $x > 0$:*

$$\sqrt{x} = 1 + \cfrac{x-1}{2 + \cfrac{x-1}{2 + \cfrac{x-1}{2 + \ddots}}}$$

Proof. Let y denote the value of the continued fraction:

$$y = 1 + \cfrac{x-1}{2 + \cfrac{x-1}{2 + \cfrac{x-1}{2 + \ddots}}}$$

Since the continued fraction is infinite and periodic, we can observe that the denominator starting from the first 2 is equal to $1 + y$. That is:

$$2 + \cfrac{x-1}{2 + \cfrac{x-1}{2 + \ddots}} = 1 + y$$

Therefore, we can rewrite y as:

$$y = 1 + \frac{x-1}{1+y}$$

Multiplying both sides by $(1 + y)$:

$$\begin{aligned} y(1+y) &= (1+y) + (x-1) \\ y + y^2 &= 1 + y + x - 1 \\ y^2 &= x \end{aligned}$$

Since y represents the value of a continued fraction with positive terms and $x > 0$, we have $y > 0$. Therefore:

$$y = \sqrt{x}$$

This proves that the continued fraction equals $\sqrt{x}$. □

6 Approximation of rational numbers

todo

7 Relation to Pell's Equation

todo

Theorem 7.1 (Pell's Equation Relation)**.** *If p, q are integers such that $p^2 - n \cdot q^2 = \pm 1$, Then p, q are convergents of $\sqrt{n}$.*

8 Solving linear Diophantine Equations

todo

9 Gauss's continued fraction

In 1813, Carl Friedrich Gauss published a celebrated paper [2] introducing what is now known as **Gauss's continued fraction**. Gauss established that the ratio of two contiguous hypergeometric functions can be expressed as an infinite continued fraction. Because the hypergeometric function encompasses a remarkably vast array of elementary and transcendental functions as special or limiting cases, Gauss's continued fraction serves as a master generating machine for continued fraction representations of logarithms, trigonometric and inverse trigonometric functions, exponentials, and powers.

9.1 Hypergeometric Functions and Contiguous Relations

Definition 9.1 (Gaussian Hypergeometric Function)**.** *For complex parameters a, b, c with $c \notin \{0, -1, -2, \dots\}$, the **Gaussian hypergeometric function** ${}_2F_1(a, b; c; z)$ is defined for $|z| < 1$ by the power series:*

$$ {}_2F_1(a, b; c; z) = \sum_{n=0}^{\infty} \frac{(a)_n (b)_n}{(c)_n} \frac{z^n}{n!} $$

where $(q)_n$ denotes the rising factorial (or Pochhammer symbol):

$$ (q)_0 = 1, \qquad (q)_n = q(q+1)(q+2)\cdots(q+n-1) = \frac{\Gamma(q+n)}{\Gamma(q)} \quad (n \geq 1) $$

For $|z| \geq 1$, the function is defined by analytic continuation along the cut plane $\mathbb{C} \setminus [1, \infty)$.

Two hypergeometric functions are said to be **contiguous** if their parameters differ by integers of magnitude ± 1. Gauss discovered that any hypergeometric function can be linearly expressed in terms of any two functions contiguous to it. To construct the continued fraction, two fundamental contiguous relations are required.

Lemma 9.2 (Gauss's Contiguous Relations)**.** *Let* $f(a,b;c;z) = {}_2F_1(a,b;c;z)$. *Then the following two recurrence identities hold:*

$$f(a,b;c;z) - f(a,b+1;c+1;z) = -\frac{a(c-b)}{c(c+1)} z \cdot f(a+1,b+1;c+2;z) \tag{1}$$

$$f(a,b+1;c+1;z) - f(a+1,b+1;c+2;z) = -\frac{(b+1)(c-a+1)}{(c+1)(c+2)} z \cdot f(a+1,b+2;c+3;z) \tag{2}$$

Proof. We prove identity (1) by comparing the coefficients of z^n on both sides.

Let the series expansion of the left-hand side of (1) be written as $\sum_{n=0}^{\infty} C_n z^n$. For $n = 0$:

$$C_0 = 1 - 1 = 0$$

For $n \geq 1$, the coefficient of z^n is:

$$C_n = \frac{(a)_n (b)_n}{(c)_n n!} - \frac{(a)_n (b+1)_n}{(c+1)_n n!}$$

Factoring out $\frac{(a)_n}{n!}$:

$$C_n = \frac{(a)_n}{n!} \left[\frac{(b)_n}{(c)_n} - \frac{(b+1)_n}{(c+1)_n} \right]$$

Recall that $(x+1)_n = \frac{x+n}{x}(x)_n$ for any $x \neq 0$. Applying this to $(b+1)_n$ and $(c+1)_n$, we have:

$$(b+1)_n = \frac{b+n}{b}(b)_n, \qquad (c+1)_n = \frac{c+n}{c}(c)_n$$

Substituting these into the bracketed term:

$$\begin{aligned}
\frac{(b)_n}{(c)_n} - \frac{(b+1)_n}{(c+1)_n} &= \frac{(b)_n}{(c)_n} \left[1 - \frac{c(b+n)}{b(c+n)} \right] \\
&= \frac{(b)_n}{(c)_n} \left[\frac{b(c+n) - c(b+n)}{b(c+n)} \right] \\
&= \frac{(b)_n}{(c)_n} \left[\frac{bc + bn - bc - cn}{b(c+n)} \right] \\
&= \frac{(b)_n}{(c)_n} \cdot \frac{n(b-c)}{b(c+n)} \\
&= -\frac{(c-b)}{c+n} \cdot \frac{n(b)_n}{b(c)_n}
\end{aligned}$$

Notice that:

$$\frac{(b)_n}{b} = (b+1)_{n-1}, \qquad (a)_n = a(a+1)_{n-1}, \qquad \frac{n}{n!} = \frac{1}{(n-1)!}$$

and

$$(c)_n (c+n) = (c)_{n+1} = c(c+1)_n = c(c+1)(c+2)_{n-1}$$

Multiplying these terms together yields:

$$C_n = -\frac{a(c-b)}{c(c+1)} \cdot \frac{(a+1)_{n-1}(b+1)_{n-1}}{(c+2)_{n-1}(n-1)!}$$

Therefore, summing over all $n \geq 1$:

$$\begin{aligned}\sum_{n=1}^{\infty} C_n z^n &= -\frac{a(c-b)}{c(c+1)} z \sum_{n=1}^{\infty} \frac{(a+1)_{n-1}(b+1)_{n-1}}{(c+2)_{n-1}(n-1)!} z^{n-1} \\ &= -\frac{a(c-b)}{c(c+1)} z \sum_{m=0}^{\infty} \frac{(a+1)_m (b+1)_m}{(c+2)_m m!} z^m \\ &= -\frac{a(c-b)}{c(c+1)} z \cdot f(a+1, b+1; c+2; z)\end{aligned}$$

which establishes identity (1).

To prove identity (2), we use the symmetry of the hypergeometric function with respect to its first two parameters:

$$f(a, b+1; c+1; z) = f(b+1, a; c+1; z)$$

Applying identity (1) with parameters $(A, B, C) = (b+1, a, c+1)$, we obtain:

$$\begin{aligned}f(b+1, a; c+1; z) - f(b+1, a+1; c+2; z) &= -\frac{A(C-B)}{C(C+1)} z \cdot f(A+1, B+1; C+2; z) \\ &= -\frac{(b+1)(c+1-a)}{(c+1)(c+2)} z \cdot f(b+2, a+1; c+3; z)\end{aligned}$$

Using symmetry once more on the functions, $f(b+1, a+1; c+2; z) = f(a+1, b+1; c+2; z)$ and $f(b+2, a+1; c+3; z) = f(a+1, b+2; c+3; z)$, which directly produces identity (2). □

9.2 Derivation of Gauss's Continued Fraction

We now assemble the contiguous relations into an alternating sequence of hypergeometric functions. Define the sequence of functions $\{f_n(z)\}_{n=0}^{\infty}$ by:

$$\begin{aligned}f_{2k}(z) &= {}_2F_1(a+k, b+k; c+2k; z) \qquad (k \geq 0) \\ f_{2k+1}(z) &= {}_2F_1(a+k, b+k+1; c+2k+1; z) \qquad (k \geq 0)\end{aligned}$$

Thus the first few terms of the sequence are:

$$\begin{aligned}f_0(z) &= {}_2F_1(a, b; c; z) \\ f_1(z) &= {}_2F_1(a, b+1; c+1; z) \\ f_2(z) &= {}_2F_1(a+1, b+1; c+2; z) \\ f_3(z) &= {}_2F_1(a+1, b+2; c+3; z) \\ f_4(z) &= {}_2F_1(a+2, b+2; c+4; z)\end{aligned}$$

Observe that applying Lemma 9.2 to $(a+k, b+k, c+2k)$ gives:

$$f_{2k}(z) - f_{2k+1}(z) = -d_{2k+1} z \cdot f_{2k+2}(z)$$

where

$$d_{2k+1} = \frac{(a+k)(c-b+k)}{(c+2k)(c+2k+1)} \qquad (k \geq 0)$$

Similarly, applying identity (2) with shifted parameters gives:

$$f_{2k-1}(z) - f_{2k}(z) = -d_{2k} z \cdot f_{2k+1}(z)$$

where

$$d_{2k} = \frac{(b+k)(c-a+k)}{(c+2k-1)(c+2k)} \qquad (k \geq 1)$$

In general, for all $n \geq 0$, the sequence satisfies the uniform three-term recurrence relation:

$$f_n(z) - f_{n+1}(z) = -d_{n+1} z \cdot f_{n+2}(z) \tag{3}$$

Theorem 9.3 (Gauss's Continued Fraction)**.** *Let $a, b, c \in \mathbb{C}$ with $c \notin \{0, -1, -2, \dots\}$. Then the ratio of contiguous hypergeometric functions admits the continued fraction expansion:*

$$\frac{{}_2F_1(a, b+1; c+1; z)}{{}_2F_1(a, b; c; z)} = \cfrac{1}{1 - \cfrac{d_1 z}{1 - \cfrac{d_2 z}{1 - \cfrac{d_3 z}{1 - \ddots}}}}$$

where the coefficients $\{d_n\}_{n=1}^{\infty}$ are given by:

$$d_{2k+1} = \frac{(a+k)(c-b+k)}{(c+2k)(c+2k+1)} \qquad (k \geq 0)$$
$$d_{2k} = \frac{(b+k)(c-a+k)}{(c+2k-1)(c+2k)} \qquad (k \geq 1)$$

Proof. Dividing the uniform three-term recurrence (3) by $f_{n+1}(z)$ (assuming $f_{n+1}(z) \neq 0$), we obtain:

$$\frac{f_n(z)}{f_{n+1}(z)} - 1 = -d_{n+1} z \cdot \frac{f_{n+2}(z)}{f_{n+1}(z)}$$

which can be rearranged as:

$$\frac{f_n(z)}{f_{n+1}(z)} = 1 - \frac{d_{n+1} z}{\dfrac{f_{n+1}(z)}{f_{n+2}(z)}}$$

Taking the reciprocal of both sides yields:

$$\frac{f_{n+1}(z)}{f_n(z)} = \frac{1}{1 - \dfrac{d_{n+1} z}{\dfrac{f_{n+1}(z)}{f_{n+2}(z)}}} \tag{4}$$

We now proceed by repeatedly substituting (4) into itself for successive values of n. For $n = 0$:

$$\frac{f_1(z)}{f_0(z)} = \cfrac{1}{1 - \cfrac{d_1 z}{\cfrac{f_1(z)}{f_2(z)}}}$$

Substituting the expression for $\frac{f_1(z)}{f_2(z)}$ from $n = 1$:

$$\frac{f_1(z)}{f_0(z)} = \cfrac{1}{1 - \cfrac{d_1 z}{1 - \cfrac{d_2 z}{\cfrac{f_2(z)}{f_3(z)}}}}$$

Continuing this process inductively for N steps:

$$\frac{f_1(z)}{f_0(z)} = \cfrac{1}{1 - \cfrac{d_1 z}{1 - \cfrac{d_2 z}{\ddots - \cfrac{d_N z}{\cfrac{f_N(z)}{f_{N+1}(z)}}}}}$$

In the cut complex plane $\mathbb{C}\backslash[1, \infty)$, the remainder term satisfies $\lim_{N\to\infty} \frac{f_{N+1}(z)}{f_N(z)} = 1$ for $|z| < 1$, ensuring convergence of the continued fraction to the meromorphic function $\frac{f_1(z)}{f_0(z)}$ [3]. Substituting the explicit formulas for d_{2k+1} and d_{2k} completes the proof. □

9.3 Special Cases and Applications

Gauss's continued fraction is remarkable because specializing the parameters a, b, c and the variable z yields continued fraction expansions for a large catalog of classical functions. Below we examine several key applications.

9.3.1 The Natural Logarithm

Recall that the Taylor series of $\ln(1 + z)$ around $z = 0$ is:

$$\ln(1 + z) = \sum_{k=1}^{\infty} \frac{(-1)^{k-1} z^k}{k} = z - \frac{z^2}{2} + \frac{z^3}{3} - \dots$$

Notice that the hypergeometric function with parameters $a = 1, b = 1, c = 2$ evaluated at $-z$ is:

$$\begin{aligned} {}_2F_1(1, 1; 2; -z) &= \sum_{n=0}^{\infty} \frac{(1)_n (1)_n}{(2)_n n!} (-z)^n = \sum_{n=0}^{\infty} \frac{n!\, n!}{(n+1)!\, n!} (-1)^n z^n \\ &= \sum_{n=0}^{\infty} \frac{(-1)^n z^n}{n+1} = \frac{\ln(1+z)}{z} \end{aligned}$$

Now set $a = 1, b = 0, c = 1$ in Theorem 9.3. Since $(0)_0 = 1$ and $(0)_n = 0$ for $n \geq 1$, we have:

$$ {}_2F_1(1, 0; 1; -z) = 1, \qquad {}_2F_1(1, 1; 2; -z) = \frac{\ln(1+z)}{z} $$

Hence the ratio is:

$$ \frac{{}_2F_1(1, 1; 2; -z)}{{}_2F_1(1, 0; 1; -z)} = \frac{\ln(1+z)}{z} $$

Evaluating the coefficients d_n from Theorem 9.3 with $a = 1, b = 0, c = 1$ and argument $-z$ (so $-d_n(-z) = +d_n z$):

$$ d_{2k+1} = \frac{(1+k)(1-0+k)}{(1+2k)(1+2k+1)} = \frac{(k+1)^2}{(2k+1)(2k+2)} = \frac{k+1}{2(2k+1)} \qquad (k \geq 0) $$

$$ d_{2k} = \frac{(0+k)(1-1+k)}{(1+2k-1)(1+2k)} = \frac{k^2}{2k(2k+1)} = \frac{k}{2(2k+1)} \qquad (k \geq 1) $$

The first few coefficients are:

$$ d_1 = \frac{1}{2}, \quad d_2 = \frac{1}{6}, \quad d_3 = \frac{1}{3}, \quad d_4 = \frac{1}{5}, \quad d_5 = \frac{3}{10}, \quad d_6 = \frac{3}{14} $$

Clearing the internal denominators level-by-level (by applying equivalence transformations that multiply the numerator and denominator of the nth level by n), we obtain:

Lemma 9.4 (Gauss Continued Fraction for $\ln(1+z)$)**.** *For any $z \in \mathbb{C} \setminus (-\infty, -1]$:*

$$ \ln(1+z) = \cfrac{z}{1 + \cfrac{1^2 z}{2 + \cfrac{1^2 z}{3 + \cfrac{2^2 z}{4 + \cfrac{2^2 z}{5 + \cfrac{3^2 z}{6 + \ddots}}}}}} $$

In particular, setting $z = 1$ gives an accelerated continued fraction for $\ln(2)$:

$$ \ln(2) = \cfrac{1}{1 + \cfrac{1^2}{2 + \cfrac{1^2}{3 + \cfrac{2^2}{4 + \cfrac{2^2}{5 + \cfrac{3^2}{6 + \ddots}}}}}} $$

Remark 9.1. *Compare this expansion for $\ln(2)$ with the one obtained in Sec-*

tion 4.1, Lemma 4.5:

$$\ln(2) = \cfrac{1}{1 + \cfrac{1^2}{1 + \cfrac{2^2}{1 + \cfrac{3^2}{1 + \ddots}}}}$$

While the expansion in Lemma 4.5 was derived from the slowly alternating harmonic series $\sum \frac{(-1)^{k+1}}{k}$ *and has linear partial denominators* 1*, Gauss's expansion features growing partial denominators* $1, 2, 3, 4, 5, \ldots$*, leading to dramatically faster numerical convergence.*

9.3.2 The Arctangent Function and π

Consider the series expansion of $\arctan(z)$:

$$\arctan(z) = \sum_{k=0}^{\infty} \frac{(-1)^k z^{2k+1}}{2k+1} = z - \frac{z^3}{3} + \frac{z^5}{5} - \frac{z^7}{7} + \ldots$$

Dividing by z:

$$\frac{\arctan(z)}{z} = \sum_{k=0}^{\infty} \frac{(-1)^k (z^2)^k}{2k+1} = \sum_{k=0}^{\infty} \frac{(1/2)_k (1)_k}{(3/2)_k k!} (-z^2)^k = {}_2F_1\left(\frac{1}{2}, 1; \frac{3}{2}; -z^2\right)$$

We set $a = 1/2, b = 0, c = 1/2$ and replace z with $-z^2$ in Theorem 9.3. Then:

$${}_2F_1\left(\frac{1}{2}, 0; \frac{1}{2}; -z^2\right) = 1, \qquad {}_2F_1\left(\frac{1}{2}, 1; \frac{3}{2}; -z^2\right) = \frac{\arctan(z)}{z}$$

The coefficients d_n are given by:

$$d_{2k+1} = \frac{(1/2+k)(1/2-0+k)}{(1/2+2k)(1/2+2k+1)} = \frac{(2k+1)^2}{(4k+1)(4k+3)} \qquad (k \geq 0)$$

$$d_{2k} = \frac{(0+k)(1/2-1/2+k)}{(1/2+2k-1)(1/2+2k)} = \frac{(2k)^2}{(4k-1)(4k+1)} \qquad (k \geq 1)$$

Clearing internal denominators yields:

Lemma 9.5 (Gauss Continued Fraction for $\arctan(z)$)**.** *For* $z \in \mathbb{C} \setminus \{iy : |y| \geq 1\}$*:*

$$\arctan(z) = \cfrac{z}{1 + \cfrac{1^2 z^2}{3 + \cfrac{2^2 z^2}{5 + \cfrac{3^2 z^2}{7 + \cfrac{4^2 z^2}{9 + \ddots}}}}}$$

Evaluating at $z = 1$, since $\arctan(1) = \frac{\pi}{4}$, we obtain:

$$\frac{\pi}{4} = \cfrac{1}{1 + \cfrac{1^2}{3 + \cfrac{2^2}{5 + \cfrac{3^2}{7 + \cfrac{4^2}{9 + \ddots}}}}}$$

Remark 9.2. *This formula for $\pi/4$ offers an instructive contrast with the continued fraction obtained in Lemma 4.7:*

$$\frac{\pi}{4} = \cfrac{1}{1 + \cfrac{1^2}{2 + \cfrac{3^2}{2 + \cfrac{5^2}{2 + \ddots}}}}$$

In Gauss's formula, the partial numerators are the squares of consecutive integers $1^2, 2^2, 3^2, 4^2, \ldots$ and the partial denominators are the odd integers $3, 5, 7, 9, \ldots$, whereas the alternating series conversion of Section 4.2 features the odd squares in the numerators and constant 2 in the denominators.

9.3.3 Confluent Limits: The Exponential and Tangent Functions

Gauss's continued fraction also extends to confluent hypergeometric functions by taking suitable limits as parameters grow without bound. Replacing z by z/b and taking the limit as $b \to \infty$, the hypergeometric function ${}_2F_1(a, b; c; z/b)$ tends to Kummer's confluent hypergeometric function ${}_1F_1(a; c; z)$. Applying this limit to Gauss's continued fraction produces corresponding continued fractions for ${}_1F_1(a; c; z)$.

An immediate and famous consequence is Lambert's continued fraction for the tangent function. Using the relation $\tan(z) = \frac{1}{i}\tanh(iz)$ and the continued fraction for the ratio of confluent hypergeometric functions (or directly from the hypergeometric form of $\tan(z)$), one arrives at:

Lemma 9.6 (Lambert's Continued Fraction for $\tan(z)$)**.** *For any $z \in \mathbb{C}$ where $\tan(z)$ is defined:*

$$\tan(z) = \cfrac{z}{1 - \cfrac{z^2}{3 - \cfrac{z^2}{5 - \cfrac{z^2}{7 - \ddots}}}}$$

Equivalently, for the hyperbolic tangent:

$$\tanh(z) = \cfrac{z}{1 + \cfrac{z^2}{3 + \cfrac{z^2}{5 + \cfrac{z^2}{7 + \ddots}}}}$$

Remark 9.3. *As mentioned in Section 1, Johann Heinrich Lambert used this very continued fraction in 1761 to provide the first rigorous proof that π is irrational. Lambert proved that if x is a non-zero rational number, then $\tan(x)$ cannot be rational. Since $\tan(\pi/4) = 1$ is rational, $\pi/4$ (and hence π) must be irrational. Gauss's continued fraction provides the unified analytic framework from which Lambert's continued fraction naturally descends.*

10 General continued fraction examples

todo

10.1 pi

todo

References

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[3] Hubert Stanley Wall. *Analytic Theory of Continued Fractions.* D. Van Nostrand Company, Inc., New York, 1948.